Wiley
Modelling and Simulation in Engineering
Volume 2017, Article ID 6720781, 11 pages
https://doi.org/10.1195/2024/4922781

Research Article

Hybrid Crosswind-Vortex Augmentation for Continuous Electricity Generation in Solar Towers

Arun Mahajan[1], Sirirat Chaisiri[2], Youssef H. El-Masry[3], Qi Yan[4], and Markus Schwarz[4]

[1]*Department of Energy Engineering, King Mongkut's University of Technology North Bangkok, Thailand*
[2]*Department of Renewable Energy, Alexandria University, Egypt*
[3]*Institute for Clean Energy, Shanghai Jiao Tong University, China*
[4]*Department of Mechanical and Process Engineering, University of Duisburg-Essen, Germany*

Received 19 February 2017; Revised 19 September 2017; Accepted 5 October 2017

Academic Editor: Omer Alawi

Abstract

This research presents an improvement to the traditional solar updraft tower, which relies solely on solar energy and cannot operate continuously throughout the day. The enhancement involves a hybrid energy approach by installing a vortex generator at the top of the tower to convert crosswinds into a vortex flow at the chimney's top. This modification induces an updraft within the tower, enabling it to generate electricity continuously, even at night when there is no sunlight. The aim is to enable the solar updraft tower to harness crosswind energy without altering the tower's main structure. This involves developing a vortex generator from a unidirectional wind intake design to a three-directional intake, enhancing the feasibility of commercial installation. Additionally, various designs and heights of vortex generators were developed, considering different crosswind speeds (2, 4, 6, and 8 m/s). The research utilizes the finite element method, along with real model construction, to validate the reliability of the study's findings. The results indicate that the updraft speed is directly proportional to the crosswind speed. From a physical standpoint, the vortex generator with a height equal to D produced the best results in all experiments. The square, cylindrical, and diffuser shapes increased the wind speed inside the chimney by 60%, 41%, and 48%, respectively. These results from various shapes provide effective design and development guidelines for the future commercial use of vortex generators.

Keywords: crosswind; solar updraft tower; suctioned updraft; vortex generator; wind energy

1. Introduction

The main system of an original solar updraft tower consists of three main components that help transfer the heat produced by the sun into electricity: a collector, a cylindrical tower, and a turbine. The collector is a transparent roof used to accumulate heat or high-temperature air from the sun. The collected heat from the collector is then turned into an updraft, which is directed towards the tower outlet. The updraft wind then drives the chimney turbine at the outlet, producing electricity.

Many previous studies have developed the solar updraft tower system based on its three main components. In 2005, Schlaich et al. [1] designed and constructed a prototype to address technical issues for future commercial use. Subsequent studies [2, 3] focused on refining electricity production technology and resolving construction challenges. Recent studies have employed computational fluid dynamics models to enhance the tower's efficiency [4, 5], with some proposing structural improvements such as transitioning from a cylindrical to a diffuser-shaped tower to optimize thermal updraft speed [6–9]. Others explored the tower's structural flexibility [10, 11], an angled collector hood [12], and various turbine configurations [13].

Some studies have focused on the impact of crosswinds on the electricity production of solar updraft towers. Study [14] specifically examined the effects of crosswinds at both the tower's chimney outlet and collector inlet. It found that low-temperature crosswinds entering the collector's inlet reduced the buoyant driving force of the updraft wind by altering heat transmission. Conversely, crosswinds at the outlet positively influenced the tower's operation by enhancing buoyant airflow within the structure. While certain studies [15] highlighted the disadvantages of crosswinds, others explored how crosswinds could optimize tower efficiency. Study [16] demonstrated that crosswinds could lower the tower outlet temperature, thereby potentially increasing electricity production. This finding was corroborated by the results presented in [17].

While most studies focused on altering the tower's fundamental structure, some researchers explored enhancing

its efficiency without modifying its core components [18, 19]. These studies proposed attaching an accessory known as a vortex generator to the tower. The vortex generator operates by transforming the crosswind at the outlet into a vortex, thereby enhancing the suction effect of the updraft, as illustrated in Figure 1. Both used the tower at D=320 mm in diameter and the height at h=2000mm. The generator, a single square-shaped windplate, shared the same width as the tower outlet, while its height h_{vg} was determined as $D/2$ and D, respectively.

This research presents an approach to harnessing crosswind currents by developing a vortex generator to enhance the feasibility of commercial applications. To capture wind from all directions, a three-directional vortex generator is specified for installation on traditional solar updraft towers without modifying the tower's main structure. The research is divided into two main parts: numerical simulation and experimental investigation. Numerical simulation helps reduce investigation time and costs. If accurate, it can guide the design of vortex generators without the need for physical experiments. However, the numerical model must be validated to ensure the reliability of the study's results.

2. Methodology and Equipment

2.1. A Physical Experiment. This study utilized a low-speed wind tunnel to validate results obtained from the tower equipped with various types of vortex generators. The wind tunnel dimensions were 800mm in height and width and 2000 mm in length with a maximum wind velocity of 12 m/s. To simulate the velocity differences observed at different heights within the tower, the wind was introduced exclusively at the top section of the tower, as depicted in Figure 2. Typically, wind speed decreases near surfaces due to shear stress. This lower velocity caused vertical instability in the crosswind as it descended towards the floor, resulting in higher wind speeds at the top of the chimney compared to the bottom.

This study replicated the model conditions used in Study [19], downsizing at a ratio of 1:3.2 due to technical limitations of the wind tunnel size and its wind-generation efficiency. The experiments were conducted at wind speeds of 2, 4, 6, and 8m/s. The selected wind speed range represents the average wind speed typically observed throughout the year in Thailand. The tower used in this study was 100 mm in diameter D and 625mm in height h.

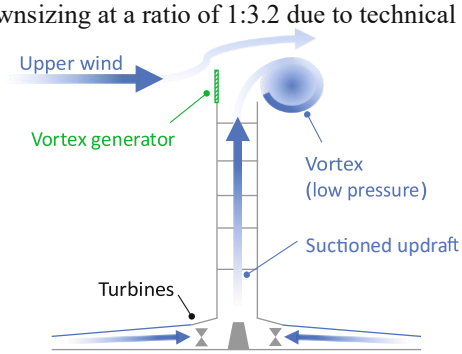


Figure 1: The principle of the vortex generator involves transforming the crosswind at the tower outlet into a vortex.

The single square-plate vortex generator in [18] was at $D/2$ and D in h_{vg} height and shared the width W_{vg} with the tower outlet's width D_{out}, shown in Figure 3(a). The three-dimensional vortex generator in various shapes, in the meantime, was at $D/2$ and D in h_{vg} height and $D/2$ in W_{vg} width, as shown in Figures 3(b), 3(c), and 3(d). The researcher installed a hot-wire anemometer to collect the suction updraft velocity (w). At 8m/s of windspeed, the researcher also used a digital manometer and a static tube to collect the static pressure (Δp_s). All instruments were installed on the wind leeside of the chimney, as depicted in Figure 2. Additionally, a smoke generator was employed in the experiment to visualize wind streamlines under wind speeds of 4m/s.

For this study, a pressure coefficient was calculated by dividing the static pressure by the dynamic pressure of the wind entering the tower, written in Equation (1).

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2} \quad (1)$$

In this equation, U_∞ means the velocity of the fluid (meters per second), p means static pressure at a pressure coefficient assessment point (kilograms per square meter), and p_∞ means static pressure of a freestream (kilograms per square meter).

2.2. A Numerical Model. This study's numerical model was generated using Ansys Fluent (commercial software), a computing program designed for fluid analysis. The advantages of this software include not only a variety of flow solvers but also multiple types of turbulence models that can be applied in current projects. Additionally, it offers high flexibility in grid customization [20]. Two calculation patterns—a schematic calculation domain format in Figure 4 and a computational grid in Figure 5—were used as the

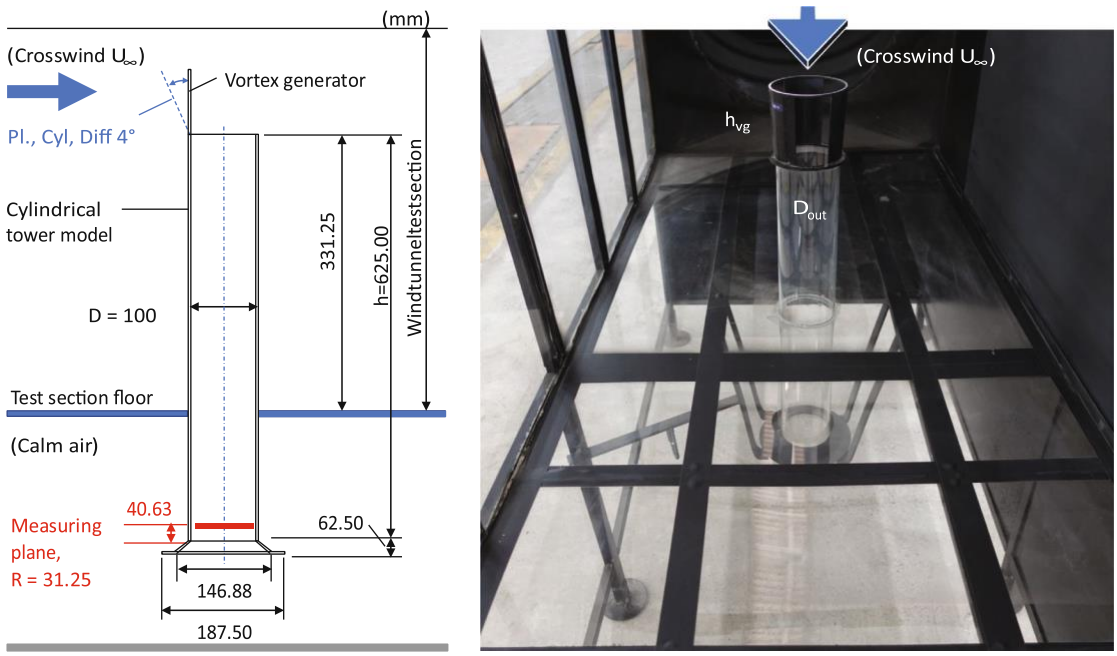


Figure 2: General conditions of the wind tunnel experiment.

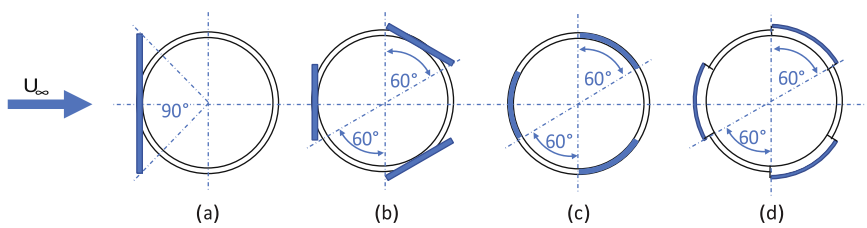
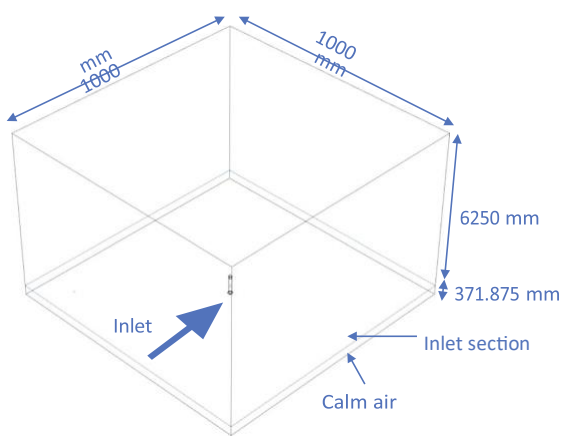


Figure 3: A top view of various vortex generators in the experiment. (a) A single-plate vortex generator in [18]. (b) A square-plate vortex generator. (c) A cylindrical-shape vortex generator. (d) A 4° diffuser-shaped vortex generator.



4: A schematic of the calculation domain.

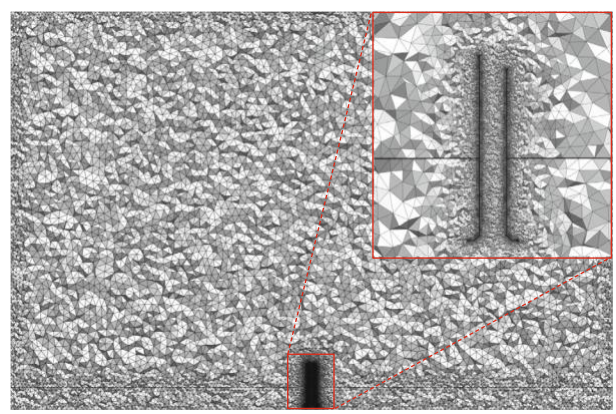


Figure 5: The computational grid. Figure

main tools of the model. The computational grid mainly consisted of tetrahedral shapes, designed to mimic the actual experimental setup as seen in Figure 5. The domain used for the model also included mesh details suitable for checking element quality, skewness, orthogonal quality, and an appropriate wall function to ensure convergence following turbulence model processing [21–25].

This study used Launder and Spalding's [26] standard k–ε turbulence model. The model is computationally efficient while maintaining high accuracy [27], making it suitable for this research, which requires the creation of numerous

numerical models to find appropriate geometries and physical properties for the vortex generator. The assumptions of this research include incompressible flow, no heat transfer, and a steady state. In this numerical setting, turbulence kinetic energy (k) can be written as in Equation (2).

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + 2\mu_t E_{ij} E_{ij} - \rho\varepsilon \quad (2)$$

A decrease in the specific kinetic energy dissipation rate (ε) is shown in Equation (3).

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \frac{\partial(\rho\varepsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + C_{1\varepsilon}\frac{\varepsilon}{k}2\mu_t E_{ij}E_{ij} - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} \quad (3)$$

An equation for the Eddy viscosity can be written in Equation (4).

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (4)$$

In all equations, k means specific turbulent kinetic energy (m^2/s^2); ρ means fluid density (kilograms per cubic meter); u_i means fluid freestream velocity (meters per second); μ_t means Eddy viscosity (kg/m.s); σ_k, σ_ε, $C_{1\varepsilon}$, and $C_{2\varepsilon}$ are the standard $k-\varepsilon$ turbulence model constant; and ε is the specific kinetic energy dissipation rate (m^2/s^3).

3. Results and Analysis

3.1. Results From the Numerical Models. From Figure 6, it is observed that within the range of element sizes from 1.1e +07 to 1.7e+07, there is no significant change in wind speed at the measurement point. The accuracy of mesh generation and validated patterns will provide guidelines for initializing various models in future applications. The wind tunnel experiment in this study was conducted at wind speeds of 2, 4, 6, and 8m/s, consistent with the experiment in [19]. However, due to space limitations and maximum wind speed constraints, the wind tunnel in this study was downscaled to a 1:3.2 ratio compared to the tunnel used in [19]. Despite this scaling, the results from the physical experiment aligned with those from the numerical models. According to Figures 7, 8, 9, and 10, the rate of increase in updraft speed of the tower, with and without vortex generators, showed consistent trends in both methodologies. These results are summarized in Table 1.

According to Table 1, the tower without the vortex generator provided the lowest updraft wind speed, while the tower with the generators in any shape provided a higher wind speed. The updraft speed of the tower with the generators continued to increase as their heights became greater in every case, which were $D/2$ and D, respectively. After analyzing the relationship between vortex generator height and updraft speed, the researcher then focused on the shape of the generator and its practical implications. The experiment began by reviewing the assumptions made in [19], which were ideal but not commercially viable. The study is aimed at finding an alternative method for using vortex generators with solar updraft towers in practical applications.

To achieve this goal, the researcher experimented with changing the number of generator windplates from onedimensional to three-dimensional configurations. This alteration rendered the tower suitable for year-round operation, as it could effectively adjust to varying wind directions while maintaining stability postinstallation. The concept led to unavoidable decisions to decrease the width of the windplates to $D/2$. As seen in Figure 7, the D-width single-plate vortex generator in the study cannot be used year-round, although this generator shape enhances the updraft speed the most, as it can receive the wind in only one direction. Moreover, the plate length was longer than the tower outlet, which led to its lack of durability. This meant the researcher had to decrease the width of the windplates from D to $D/2$, which led to an updraft speed decrease of 22.61%. Meanwhile, there emerged a commercial opportunity for the three-dimensional square-shaped vortex generator, despite it resulting in a 22.85% decrease in updraft speed. This possibility spurred the development of various three-dimensional vortex generator shapes based on the square design, with the goal of identifying the most suitable windplate shape for the tower. The cylindricalshaped vortex generator was developed by curving the square windplates towards the tower outlet, enhancing stability during installation. However, it yielded the lowest updraft speed, despite achieving the highest velocity-increasing rate at 43.02%, as illustrated in Figure 9. Directly inspired by the study [7], the 4° diffuser-shaped vortex generator took inspiration from the diffuser-shaped tower, resulting in a speed enhancement of 49.65% at the D height. When analyzing Figures 11, 12, 13, and 14 alongside the previous message, which depicts the pressure coefficient at various points inside the tower with and without the generator, it becomes evident that the coefficient generally increases along the height of the tower. Specifically, it stabilizes at a height of 0.1m before beginning to decrease at the 0.6m mark. This increase in the coefficient was consistent across all cases. Vortex generators that enhanced updraft speed also resulted in higher pressure coefficients, as these generators lowered air pressure inside the chimney.

The data in Table 2 compares the rate of increase in updraft wind speed from numerical models in this study using an aspect ratio of 1:3.2 to the reference aspect ratio of 1:1 from [19], with identical input values. Crosswind speeds of 2, 4, 6, and 8m/s from simulations of a simple vortex generator model showed that reducing the aspect ratio led to lower updraft wind speeds at the measurement points. This trend was consistent across all cases. However, when

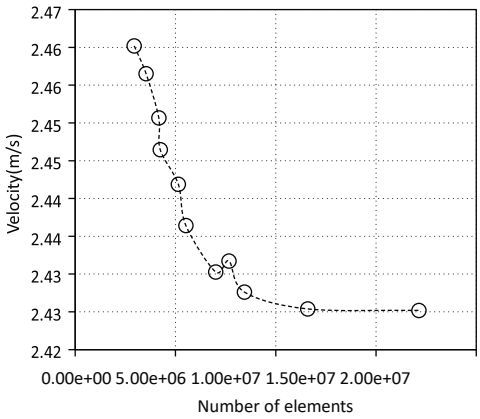


Figure 6: The number of elements required to achieve an optimum mesh.

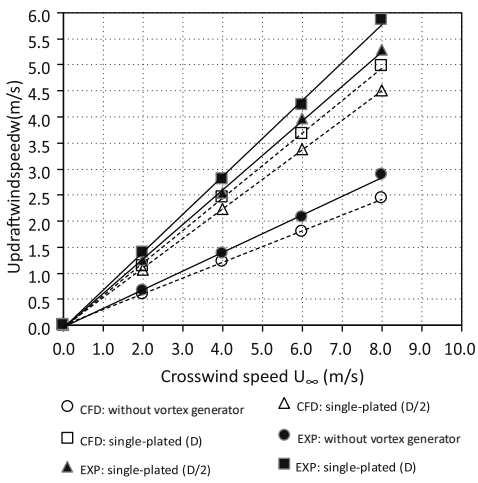


Figure 7: An updraft speed comparison between the tower without the vortex generator and with the original square-shaped vortex generator.

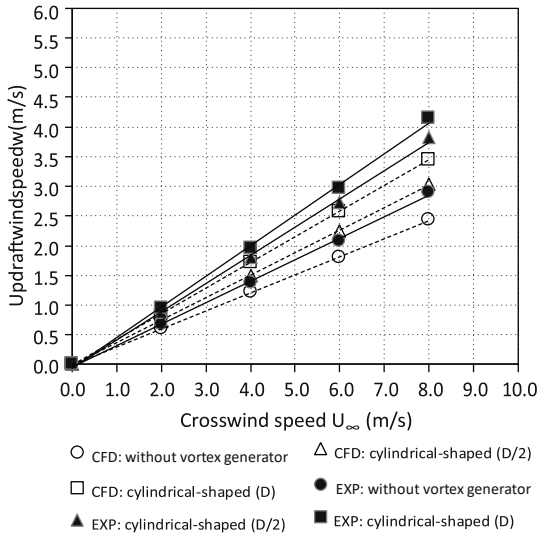


Figure 9: An updraft speed comparison between the tower without the vortex generator and with the three-dimensional cylindricalshaped vortex generator.

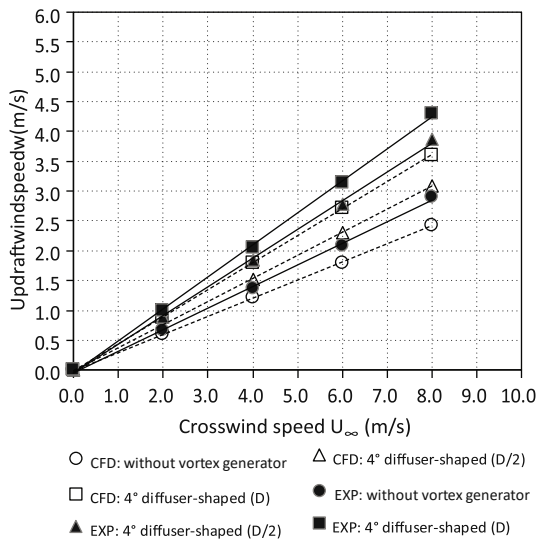


Figure 10: An updraft speed comparison between the tower without the vortex generator and with the three-dimensional 4° diffuser-shaped vortex generator.

Table 1: The increasing rate of the updraft speed of the tower with the vortex generators.

Type of vortex generator	Height (h_{vg})	Width (W_{vg})	No. of the generator windplates	Avg. updraft speed increasing rate (%)	
				EXP	CFD
Single-plated	$D/2$	D	1	86.53	83.83
Single-plated	D	D	1	104.91	100.13
Single-plated	$D/2$	$D/2$	1	—	59.16
Single-plated	D	$D/2$	1	—	79.58
Square-shaped	$D/2$	$D/2$	3	39.49	33.80
Square-shaped	D	$D/2$	3	60.31	59.24
Cylindrical-shaped	$D/2$	$D/2$	3	31.28	23.43
Cylindrical-shaped	D	$D/2$	3	43.02	41.01
Diffuser-shaped (4°)	$D/2$	$D/2$	3	34.00	26.18
Diffuser-shaped (4°)	D	$D/2$	3	49.65	48.28

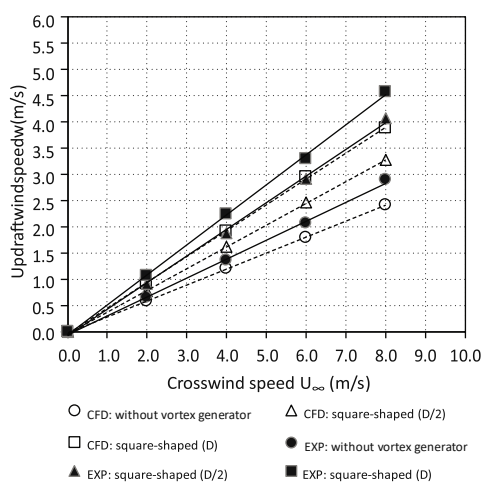
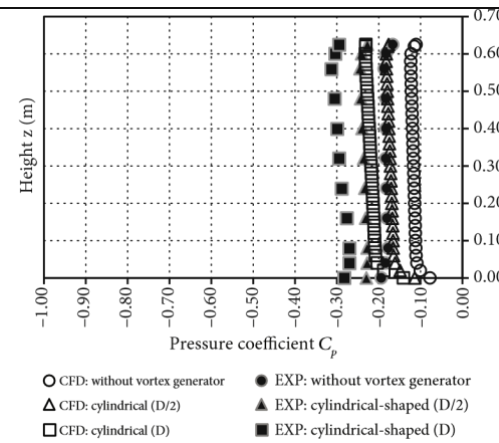


Figure 8: An updraft speed comparison between the tower without the vortex generator and with the three-dimensional square-shaped vortex generator.

comparing the rate of increase in updraft wind speed between models with and without increased wind-catching plate height, both models (aspect ratios of 1:1 and 1:3.2) showed proportional increases in updraft. The difference in updraft rate between the two models was approximately 1%–4%. Therefore, it can be concluded that reducing the aspect ratio does not significantly affect the model's input and output.



3.2. Streamline and Pressure Coefficient Results From the Vortex Generator. Figure 15 presents the results of the

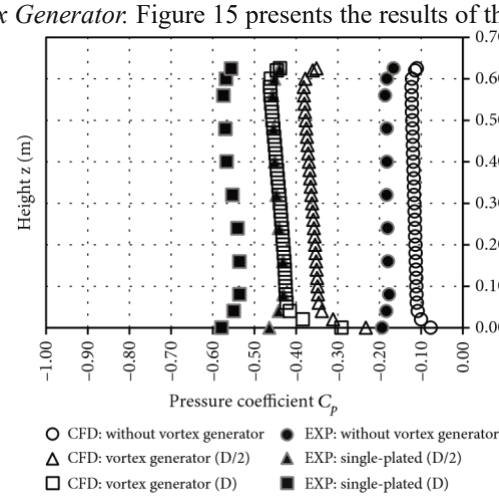


Figure 11: A pressure coefficient comparison between the cylindrical tower without the vortex generator and with the original square-shaped vortex generator.

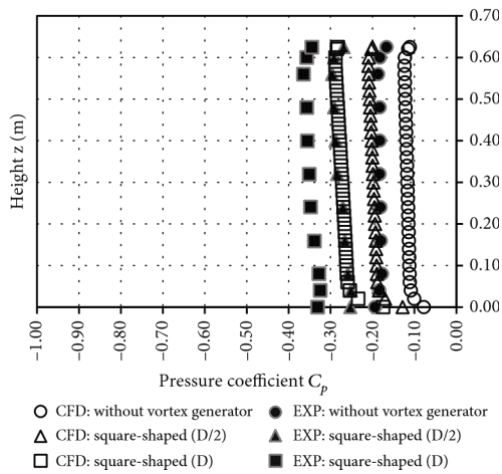


Figure 12: A pressure coefficient comparison between the cylindrical tower without the vortex generator and with the threedimensional square-shaped vortex generator.

Figure 13: A pressure coefficient comparison between the cylindrical tower without the vortex generator and the threedimensional cylindrical-shaped vortex generator.

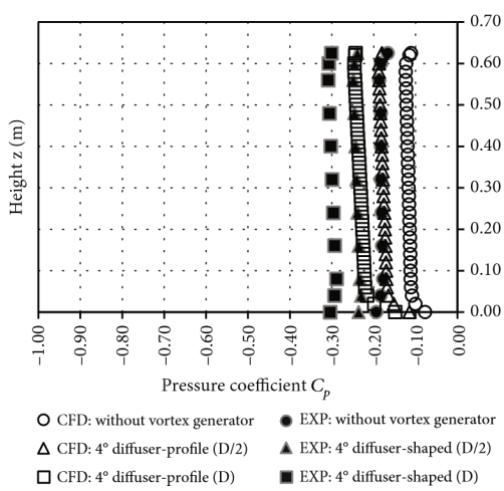


Figure 14: A pressure coefficient comparison between the cylindrical tower without the vortex generator and with the threedimensional 4° diffuser-shaped vortex generator.

Table 2: The comparison of the rate of increase in updraft wind speed of numerical models with different aspect ratios.

Types of vortex generators	Height (h_{vg})	Width (W_{vg})	No. of the generator windplates	Average updraft windspeed from CFD (w) 1:1	1:3.2
Without vortex generator	—	—	—	$0.316U_\infty$	$0.302U_\infty$
Single-plated	$D/2$	D	1	$0.590U_\infty$	$0.562U_\infty$
Single-plated	D	D	1	$0.651U_\infty$	$0.616U_\infty$
Single-plated	$D/2$	$D/2$	1	$0.495U_\infty$	$0.483U_\infty$
Single-plated	D	$D/2$	1	$0.559U_\infty$	$0.545U_\infty$

Figure 15: Comparisons of a streamlined formation between the tower without a vortex generator and with generators in various shapes at 4 m/s crosswind speed.

streamlined experiment using a smoke generator placed in the calm air area. In the streamline of the tower without the generator, the figure shows that smoke tended to flow into the inlet of the chimney before exiting through the outlet. This phenomenon occurred due to the suction effect caused by crosswinds passing through the calm air spot at the tower inlet, preventing vortex formation throughout the tower. The same phenomenon can also be seen during the experiment with the three-dimensional square-shaped generator at the $D/2$ height, in which a little amount of a streamlined spiral was seen. A visible vortex, however, can be seen at the tower outlet when the D-height squareshaped generator is installed. The vortex smoke from this type of generator had a lower density compared to the vortex smoke from the one-dimensional square-shaped generator. The visible vortex at the tower outlet can also be seen on both cylindrical-shaped generators at D and $D/2$ height. It can be seen in every case of 4° diffuser-shaped generators at every height.

When analyzing the results from the numerical models in Figure 16, it can be observed that the streamlines exhibited similar patterns in both methodologies. No vortex was

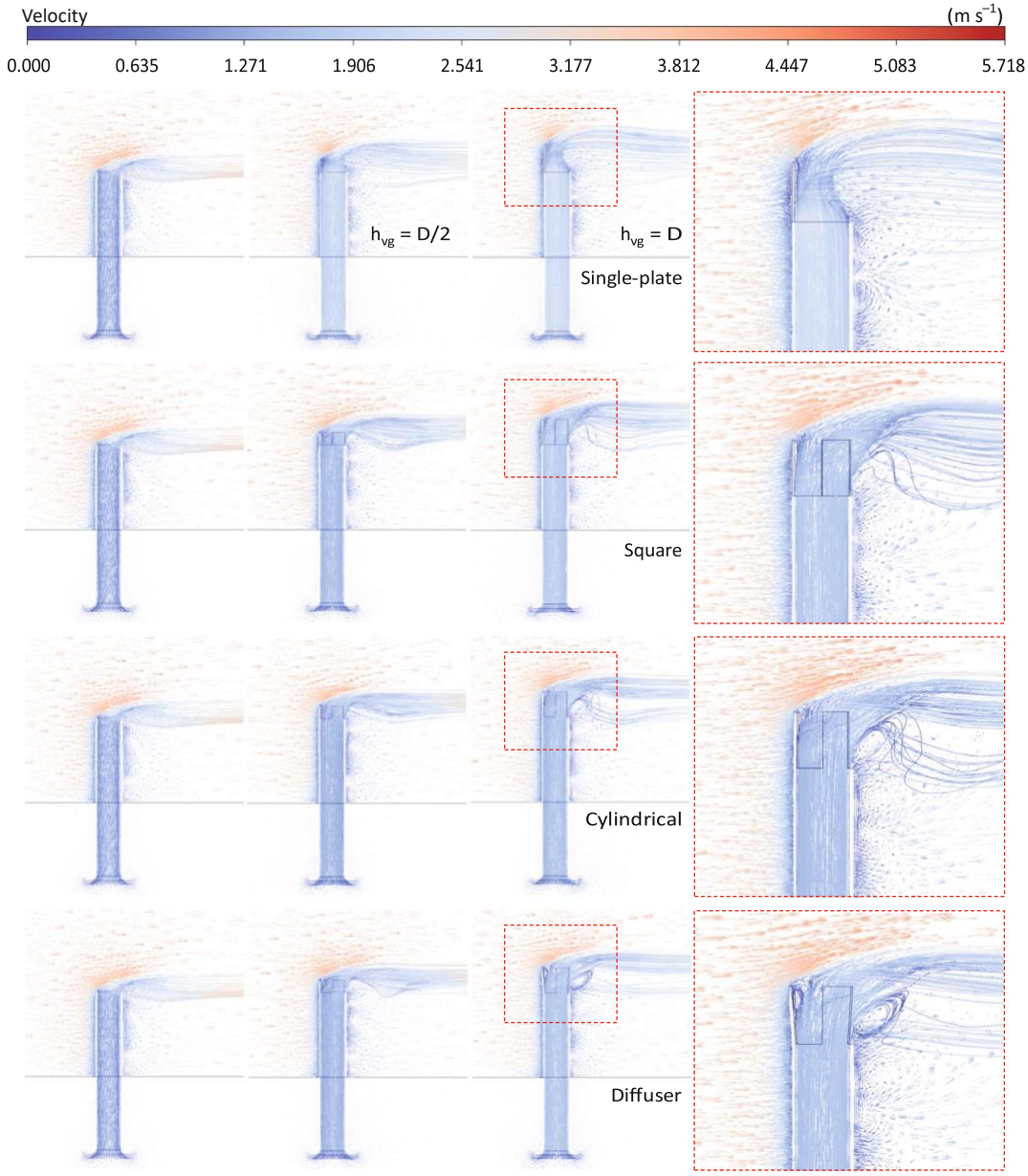


Figure 16: Numerical model comparisons on a streamlined formation between the tower without a vortex generator and with generators in various shapes at 4 m/s crosswind speed.

observed at the tower outlet for the configuration with a one-dimensional square-shaped generator, despite achieving the highest updraft speed, indicated by the lightest bluecolored line along the tower. The same phenomenon can also be seen in the three-dimensional square-shaped generator at both D and $D/2$ heights. However, some vortex formation was seen in the form of a streamline vector on the tower with the D-height square-shaped generator. A vortex formation was also seen in the D-height cylindrical-shaped generator in both streamlined and vague streamlined vector forms. The blue-colored line throughout the tower with this kind of vortex generator appeared darkest among all due to the lowest updraft speed it can generate. The tower with the D-height diffuser-shaped generator is the one that can generate the most visible vortex at the outlet, with some of those seen at the tower base as well.

The relationship between the pressure coefficient and the updraft speed was found to be inverse upon analysis. When examining Figure 16 alongside the pressure coefficient image in Figure 17, it becomes evident that the vortex area resulted in the tower having the lowest coefficient, thereby enhancing the updraft speed. The area inside the tower without any generator appeared in a lighter blue shade, indicating the

highest coefficient and consequently the lowest suction updraft at the outlet. Conversely, the coefficients were lower in towers equipped with vortex generators due to the formation of vortices from low-pressure air. This led to higher suction updrafts and increased updraft speeds at the outlet, as depicted by the darker blue areas.

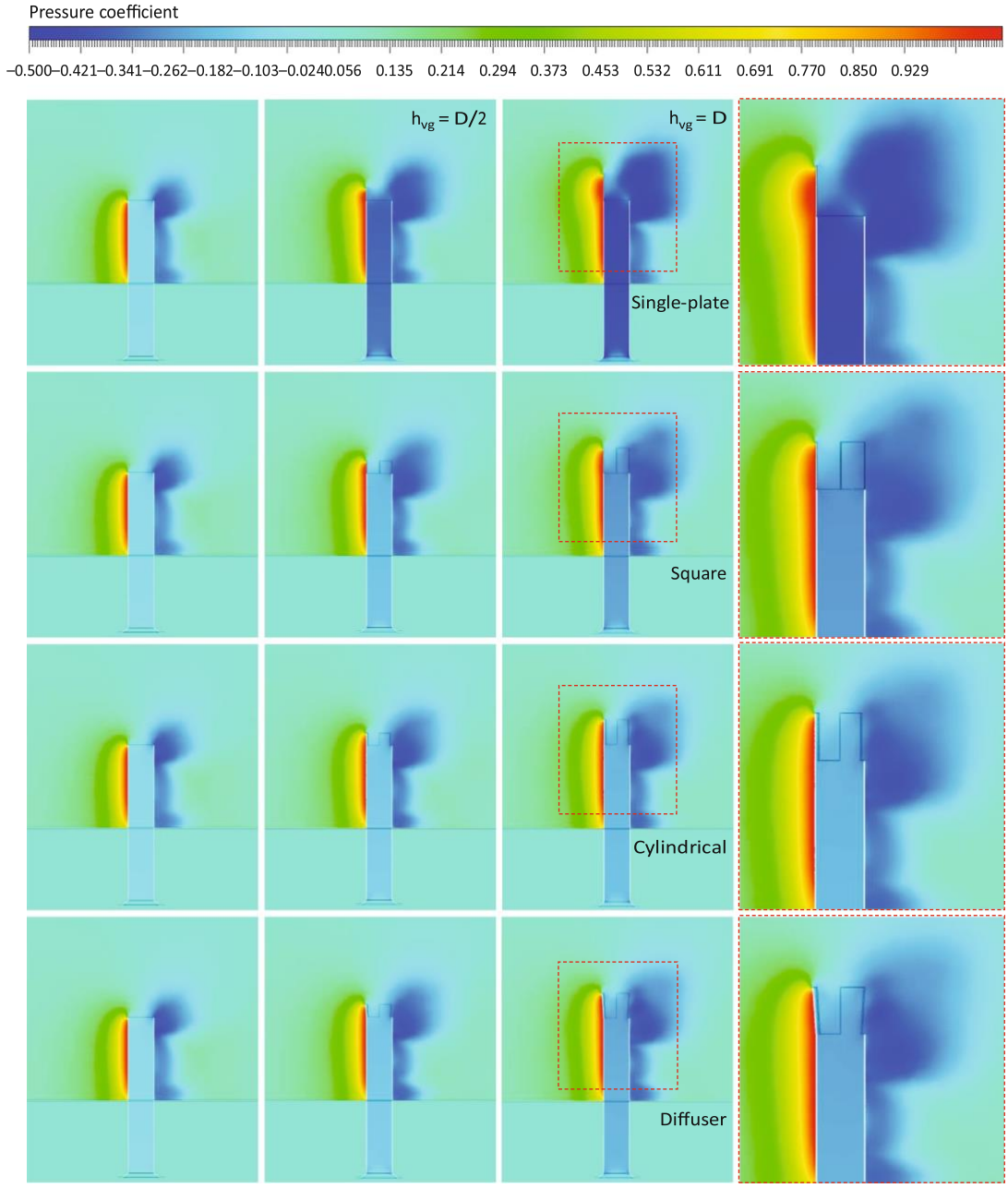


Figure 17: Comparison of pressure coefficient between the tower without a vortex generator and with generators in various shapes at 4 m/s crosswind speed.

4. Conclusion

This study focused on developing various types of vortex generators for solar updraft towers, aimed at converting crosswinds into vortices at the tower's outlet. These vortices create lower air pressure at the tower outlet, resulting in suction updrafts inside the tower. The higher updraft speeds achieved enable year-round operation. The tower utilizes sunlight during the daytime to heat the updraft and power a turbine. Conversely, crosswinds serve as the tower's primary power source at night or when sunlight is unavailable. Inspired by this principle, the study revisited the tower structure and developed a three-dimensional vortex generator aimed at commercial application with the tower. The study's conclusion can be summarized as follows:

1. The one-dimensional square-shaped generator achieveda 22.85% higher updraft wind speed

compared to the three-dimensional square-shaped generator with the same wind plate width.

2. The relationship between the updraft speed and thecrosswind velocity, as well as the width and height of the windplate, exhibited direct variation.
3. The three-dimensional vortex generator resulted in alower updraft speed due to streamlined compression between its two windplates, which retained crosswinds and affected the updraft stream exiting the outlet.
4. The three-dimensional vortex generators in squareshaped, cylindrical-shaped, and diffuser-shaped configurations enhanced the updraft speed by 60.31%, 43.02%, and 49.65%, respectively.
5. Adjusting the aspect ratio from 1:1 to 1:3.2 whilemaintaining the same crosswind speeds showed no significant change in the results. However, if larger aspect ratio differences were applied in practice, numerical simulations would still be necessary to confirm the results before installation.

The results indicated that installing the three-dimensional vortex generator can yield more favorable outcomes than the one-dimensional version, despite a lower updraft speed. This approach also enhances possibilities for commercial deployment and installation. Further studies on actual tower construction based on these findings are recommended for future research.

Nomenclature

h_{vg} height of the vortex generators (meters)
D diameter of cylindrical tower (meters)
w updraft speed (meters per second)
Δp_s static pressure difference (kilograms per square meter)
C_p pressure coefficient
ρ fluid density (kilograms per cubic meter)
U_∞ flow velocity of crosswind (meters per second)
p static pressure at the evaluated point (kilograms per square meter)
p_∞ static pressure in the freestream (kilograms per square meter)
k specific turbulent kinetic energy (m^2/s^2)
u_i incident free-stream airflow (meters per second)
μ_t turbulent or eddy viscosity (kg/m s)
σ_k constants in the conventional k-epsilon
E_{ij} component of the rate of deformation
ε turbulent kinetic energy dissipation rate (m^2/s^3)
σ_ε constants in the conventional k-epsilon
$C_{1\varepsilon}$ constants in the conventional k-epsilon
$C_{2\varepsilon}$ constants in the conventional k-epsilon

Subscripts

CFD computational fluid dynamics EXP experiment

Data Availability Statement

Data is available on request due to privacy/ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] J. Schlaich, R. Bergermann, W. Schiel, and G. Weinrebe, "Design of commercial solar updraft tower systems—utilization of solar induced convective flows for power generation," *Journal of Solar Energy Engineering*, vol. 127, no. 1, pp. 117– 124, 2005.

[2] A. Dhahri and A. Omri, "A review of solar chimney power generation technology," *International Journal of Engineering and Advanced Technology*, vol. 2, no. 3, pp. 1–17, 2013.

[3] A. Dhahri, A. Omri, and J. Orfi, "Numerical study of a solar chimney power plant," *Research Journal of Applied Sciences, Engineering and Technology*, vol. 8, no. 18, pp. 1953–1965, 2014.

[4] M. ÖNAL, A. KOÇ, Ö. KÖSE, Y. KOÇ, and H. YAĞLI, "Numerical examination of a solar chimney power plant designed for the Iskenderun region," *Konya journal of engineering sciences*, vol. 10, no. 3, pp. 548–562, 2022.

[5] M. Motoyama, K. Sugitani, Y. Ohya, T. Karasudani, T. Nagai, and S. Okada, "Improving the power generation performance of a solar tower using thermal updraft wind," *Energy and Power Engineering*, vol. 6, no. 11, pp. 362–370, 2014.

[6] Y. Ohya, M. Wataka, K. Watanabe, and T. Uchida, "Laboratory experiment and numerical analysis of a new type of solar tower efficiently generating a thermal updraft," *Energies*, vol. 9, no. 12, p. 1077, 2016.

[7] M. S. Delooee, M. R. Golzarian, and M. Tabasizadeh, "Enhancing the performance of a laboratory-scale solar chimney by geometric modifications," *Ira Conference on Solar Energy*, vol. 5, 2002.

[8] W. Shen, T. Ming, Y. Ding, Y. Wu, and R. K. de_Richter, "Numerical analysis on an industrial-scaled solar updraft power plant system with ambient crosswind," *Renewable Energy*, vol. 68, pp. 662–676, 2014.

[9] J. B. C. Silva, L. P. Gonçalves, L. Ledo, C. Maia, S. Hanriot, and J. L. Júnior, "Numerical analysis of the crosswind in small solar chimney," *International Conference on Heat Transfer*, vol. 12, pp. 301–307, 2016.

[10] Schlaich, J., Bergermann, R., Schiel, W., Weinrebe, G. (2005). Design of commercial solar updraft tower systems-utilization of solar induced convective flows for power generation. J. Sol. Energy Eng., 127(1), 117-124.

[11] Dhahri, A., Omri, A. (2013). A review of solar chimney power generation technology. Int. J. Eng. Adv. Technol., 2(3), 1-17.

[12] Dhahri, A., Omri, A., Orfi, J. (2014). Numerical study of a solar chimney power plant. Res. J. Appl. Sci. Eng. Technol., 8(18), 1953-1965.

[13] Motoyama, M., Sugitani, K., Ohya, Y., Karasudani, T., Nagai, T., Okada, S. (2014). Improving the power generation performance of a solar tower using thermal updraft wind. Energy Power Eng., 6(11), 362-370.

[14] Ohya, Y., Wataka, M., Watanabe, K., Uchida, T. (2016). Laboratory experiment and numerical analysis of a new type of solar tower efficiently generating a thermal updraft. Energies, 9(12), 1077.

[15] Shen, W., Ming, T., Ding, Y., Wu, Y., de Richter, R.K. (2014). Numerical analysis on an industrial-scaled solar updraft power plant system with ambient crosswind. Renewable Energy, 68, 662-676.

[16] Koonsrisuk, A. (2013). Comparison of conventional solar chimney power plants and sloped solar chimney power plants using second law analysis. Sol. Energy, 98, 78-84.

[17] Schlaich, J., Schiel, W., Weinrebe, G. (2006). The Solar Chimney: Electricity from the Sun. Earthscan.

[18] Lu, F., Zhang, H., Yao, X.P., et al. (2006). Review of key technologies for solar chimney power generation. East China Electric Power.

[19] Bernardes, M., Voss, A., Weinrebe, G. (2003). Thermal and technical analyses of solar chimneys. Solar Energy, 75, 511-524.

[20] Haaf, W. (1984). Solar chimneys Part II: Preliminary test results from the Manzanares pilot plant. Int. J. Solar Energy, 2, 141-161.

[21] Haaf, W., Friedrich, K., Mayr, G., Schlaich, J. (1983). Solar chimneys Part I: Principle and construction of the Manzanares pilot plant. Int. J. Solar Energy, 2, 3-20.

[22] Pretorius, J., Kr√∂ger, D. (2006). Solar chimney power plant performance. J. Sol. Energy Eng., 128(3), 302-311.

[23] Gannon, A., von Backstr√∂m, T. (2000). Solar chimney cycle analysis with system loss and solar collector performance. J. Sol. Energy Eng., 122, 133-137.

[24] Mehta, D. B., Arunkumar, H. S., Hiremath, P., & Shettar, M. (2016). Numerical Analysis on Effect of Chimney Height on Solar Updraft Power Plant. International Journal of Global Technology Initiatives, 5(1).

[25] Zhou, X., Xiao, B., Liu, W., Guo, X., Yang, J., Fan, J. (2010). Comparison of classical solar chimney power system and combined solar chimney system for power generation and seawater desalination. Desalination, 250, 249-256.

[26] N. Fatchurrohman and S. T. Chia, "Performance of hybrid nano-micro reinforced mg metal matrix composites brake calliper: simulation approach," *IOP Conference Series: Materials Science and Engineering*, vol. 257, article 012060, 2017.

[27] K. Gok, S. Inal, A. Gok, and E. Gulbandilar, "Comparison of effects of different screw materials in the triangle fixation of femoral neck fractures," *Journal of Materials Science: Materials in Medicine*, vol. 28, no. 5, p. 81, 2017.

[28] B. E. Launder and D. B. Spalding, "The numerical computation of turbulent flows," *Computer Methods in Applied Mechanics and Engineering*, vol. 3, no. 2, pp. 269–289, 1974.